

CAPSTONE PROJECT PROPOSAL

Wipro Capstone Project

Enterprise UI Automation Testing Framework using Playwright

Playwright | JavaScript | Cross-Browser Testing | HTML Reporting | POM Framework

PROJECT OVERVIEW

This project is a professional End-to-End UI Automation Testing Framework developed using Playwright for the SauceDemo (Swag Labs) e-commerce application.

The framework is designed to automate and validate critical business workflows including:

- User Authentication
- Product Catalog Validation
- Product Sorting
- Shopping Cart Operations
- Checkout Form Validation
- Order Calculation Verification
- Cross-Browser Execution

The project follows industry-standard automation practices such as Page Object Model (POM), reusable components, parallel execution, and automated reporting.

PROJECT DETAILS

Target Website : <https://www.saucedemo.com>

Framework : Playwright v1.4x+

Programming : JavaScript / TypeScript

Design Pattern : Page Object Model (POM)

Browsers Tested : Chromium, Firefox, WebKit

Reporting : Playwright HTML Report

Total Test Cases : 120+

PROJECT STRUCTURE

project/

pages/

LoginPage.js

ProductPage.js

CartPage.js

CheckoutPage.js

OrderPage.js

tests/

auth.spec.js

product.spec.js

cart.spec.js

checkout.spec.js

order.spec.js

utils/

helper.js

testData.js

playwright.config.js

package.json

README.md

SERVICES & TEST COVERAGE

1. Authentication & Security

- Valid Login
- Invalid Login
- Session Handling
- Error Message Validation

2. Product Catalog & Sorting

- Product Sorting
- Product Details Validation
- Catalog Verification

3. Cart Operations

- Add Product to Cart
- Remove Product from Cart
- Cart Badge Validation

4. Checkout Validation

- Form Validation
- Empty Field Validation
- Error Message Verification

5. Order Calculations

- Subtotal Verification
- Tax Validation
- Final Total Verification

6. Cross-Browser Testing

- Chromium Execution
- Firefox Execution
- WebKit Execution

KEY FEATURES

- Page Object Model (POM)
- Reusable Automation Components
- Cross-Browser Testing
- Parallel Test Execution
- HTML Reporting
- Screenshot Capture on Failure
- Scalable Framework Design
- Assertion-Based Validation

PLANNED TEST EXECUTION

Core Test Scenarios : 40

Browser Configurations : 3

Total Planned Tests : 120

GETTING STARTED

Prerequisites

- Node.js v18+
- npm v9+
- Visual Studio Code

INSTALLATION

`npm install`

`npx playwright install`

RUNNING TESTS

Run All Tests

`npm test`

Run Tests in Headed Mode

`npm run test:headed`

Open HTML Report

`npx playwright show-report`

CONCLUSION

This project demonstrates a scalable and enterprise-level UI Automation Testing Framework using Playwright. It showcases modern automation practices including Page Object Model (POM), reusable test architecture, automated reporting, and cross-browser execution for efficient and reliable software testing.